

College of Information Technology
Department of Computer Science
ITCS285 Database Management Systems Second
Semester 2025/2026



ITCS285 FINAL PROJECT REPORT

Name: HUSSAIN ALI H. ALI

ID: 202405120 **Section:** 05

Name: DANIA AHMED

ID: 202400485 **Section:** 05

Research supervised by Dr. Mohsin Rouached

Research Publication Tracker

Database Management System

INTRODUCTION:

1.1 PROJECT OVERVIEW:

The **Research Publication Tracker** is a database management system project designed to organize and manage university research publications. The system focuses on storing information about researchers, departments, publications, publication venues, keywords, and authorship details.

Universities produce many research outputs every year, including journal articles, conference papers, book chapters, and reports. Without a structured database, it can become difficult to track who wrote each publication, which department produced it, where it was published, and what topics it covers. This project solves that problem by designing a relational database that stores publication data in a clear, organized, and searchable way.

The main goal of this project is to apply database design concepts learned in ITCS285, including entity identification, relationships, constraints, normalization, relational schema design, SQL table creation, sample data insertion, and useful SQL query writing.

1.2 PROJECT OBJECTIVES:

- **Design** a database system for tracking *academic research publications*.
- **Store** researcher information such as *name, email, role, and department*.
- **Store** department information such as *department name and college*.

- **Store** publication information such as *title, year, type, DOI, abstract, and venue*.
- **Track** publication venues such as *journals, conferences, and publishers*.
- **Support** many-to-many relationships between *researchers and publications*.
- **Store** the order of *authors* for each *publication*.
- **Support** many-to-many relationships between *publications and keywords*.
- **Apply** *primary keys, foreign keys, unique constraints, and domain constraints*.
- **Write** SQL statements to create *tables* and insert *sample data*.
- **Write** SQL queries that produce useful reports about *publications*.

SYSTEM REQUIREMENTS:

2.1 SYSTEM SCOPE:

The system is designed as a relational database that organizes research publication data in a structured and efficient way. It reduces redundancy and allows fast retrieval of academic information through relationships between *researchers, departments, publications, and related metadata*.

Area	Description
Researchers	Stores information about faculty members, students, and external researchers.
Departments	Stores the academic department and college connected to each researcher.
Publications	Stores research output details such as title, year, type, DOI, and abstract.
Venues	Stores information about where publications are published, such as journals or conferences.
Keywords	Stores topic keywords that help classify and search publications.

Authorship	Connects researchers to publications and records author order.
Publication Keywords	Connects publications to their related keywords.

2.2 REQUIREMENT GATHERING:

These requirements were identified based on the need to manage academic research data in a university environment in a structured manner:

- **Researchers** have an ID, full name, email, role, and department.
- **Departments** have an ID, department name, and college.
- **Publications** have an ID, title, publication year, publication type, DOI, and abstract.
- **Venues** have an ID, venue name, venue type, and publisher.
- **Keywords** have an ID and keyword text.
- A **researcher** can author many publications, and a publication can have many researchers.
- The order of **authors** must be stored for each publication.
- A **publication** must be linked to one publication venue.
- A **publication** can have many keywords, and a keyword can belong to many publications.
- **Researcher** emails must be unique.
- **DOI** values must be unique when available.
- The system should support searching **publications** by researcher, department, venue, year, or keyword.
- The system should support generating **reports** such as publications per researcher and publications per department.

ER DIAGRAM:

The ER diagram below shows the main entities in the system and how they are related to each other.

Research Publication Tracker - ER Diagram

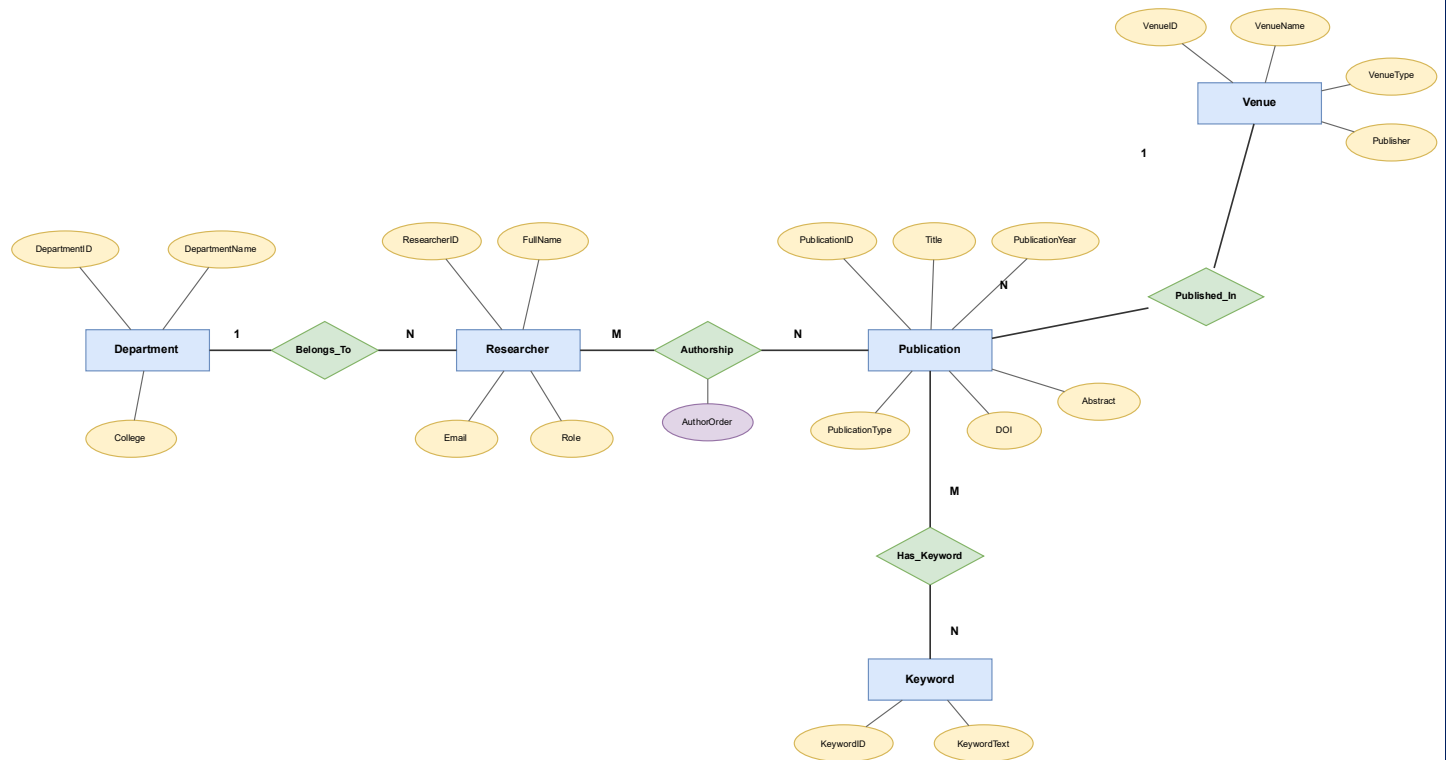


Figure 1: ER Diagram of Research Publication Tracker System

RELATIONAL SCHEMA:

The table below shows a relational database schema of the ER diagram above.

DEPARTMENT

Department(**DepartmentID**, DepartmentName, College)

Primary Key: DepartmentID

RESEARCHER

Researcher(**ResearcherID**, FullName, Email, Role, DepartmentID)

Primary Key: ResearcherID

Foreign Key: DepartmentID → Department(DepartmentID) **Unique:** Email

VENUE

Venue(**VenueID**, VenueName, VenueType, Publisher)

Primary Key: VenueID

PUBLICATION

Publication(**PublicationID**, Title, PublicationYear, PublicationType, DOI, Abstract, VenueID)

Primary Key: PublicationID

Foreign Key: VenueID → Venue(VenueID)

Unique: DOI

KEYWORD

Keyword(**KeywordID**, KeywordText)

Primary Key: KeywordID

Unique: KeywordText

AUTHORSHIP

Authorship(**ResearcherID**, **PublicationID**, AuthorOrder)

Primary Key: (ResearcherID, PublicationID) **Foreign**

Keys:

- ResearcherID → Researcher(ResearcherID)
- PublicationID → Publication(PublicationID)

PUBLICATIONKEYWORD

PublicationKeyword(**PublicationID**, **KeywordID**)

Primary Key: (PublicationID, KeywordID) **Foreign**

Keys:

- PublicationID → Publication(PublicationID)
- KeywordID → Keyword(KeywordID)

DBMS SYSTEM:

This section presents the implementation of the **Research Publication Tracker database** using **MySQL**. The relational schema was translated into SQL to create the required tables and define all relationships and constraints.

MySQL was used for this project due to its reliability and support for relational database features. The following figures show the creation of the tables and the insertion of sample data into the system. The implementation begins with the [creation of the database tables](#) using SQL statements.

```
5  -- Department table
6  * CREATE TABLE Department (
7      DepartmentID INT AUTO_INCREMENT PRIMARY KEY,
8      DepartmentName VARCHAR(100) NOT NULL,
9      College VARCHAR(100) NOT NULL
10 );
11
12  -- Researcher table
13  * CREATE TABLE Researcher (
14      ResearcherID INT AUTO_INCREMENT PRIMARY KEY,
15      FullName VARCHAR(150) NOT NULL,
16      Email VARCHAR(150) NOT NULL UNIQUE,
17      Role VARCHAR(50) NOT NULL,
18      DepartmentID INT NOT NULL,
19      FOREIGN KEY (DepartmentID) REFERENCES Department(DepartmentID),
20      CHECK (Role IN ('Faculty', 'Student', 'External Researcher'))
21 );
```

```
24 * CREATE TABLE Venue (
25     VenueID INT AUTO_INCREMENT PRIMARY KEY,
26     VenueName VARCHAR(150) NOT NULL,
27     VenueType VARCHAR(50) NOT NULL,
28     Publisher VARCHAR(150),
29     CHECK (VenueType IN ('Journal', 'Conference', 'Book', 'Report', 'Other'))
30 );
31
32  -- Publication table
33  * CREATE TABLE Publication (
34      PublicationID INT AUTO_INCREMENT PRIMARY KEY,
35      Title VARCHAR(255) NOT NULL,
36      PublicationYear INT NOT NULL,
37      PublicationType VARCHAR(50) NOT NULL,
38      DOI VARCHAR(100) UNIQUE,
39      Abstract TEXT,
40      VenueID INT NOT NULL,
41      FOREIGN KEY (VenueID) REFERENCES Venue(VenueID),
42      CHECK (PublicationYear BETWEEN 1900 AND 2025),
43      CHECK (PublicationType IN ('Journal', 'Conference', 'Book Chapter', 'Report'))
44 );
```



```

46  -- Keyword table
47  * CREATE TABLE Keyword (
48      KeywordID INT AUTO_INCREMENT PRIMARY KEY,
49      KeywordText VARCHAR(100) NOT NULL UNIQUE
50  );
51
52  -- Authorship table
53  * CREATE TABLE Authorship (
54      ResearcherID INT NOT NULL,
55      PublicationID INT NOT NULL,
56      AuthorOrder INT NOT NULL,
57      PRIMARY KEY (ResearcherID, PublicationID),
58      FOREIGN KEY (ResearcherID) REFERENCES Researcher(ResearcherID),
59      FOREIGN KEY (PublicationID) REFERENCES Publication(PublicationID),
60      CHECK (AuthorOrder > 0)
61  );
62
63  -- PublicationKeyword table
64  * CREATE TABLE PublicationKeyword (
65      PublicationID INT NOT NULL,
66      KeywordID INT NOT NULL,
67      PRIMARY KEY (PublicationID, KeywordID),
68      FOREIGN KEY (PublicationID) REFERENCES Publication(PublicationID),
69      FOREIGN KEY (KeywordID) REFERENCES Keyword(KeywordID)
70  );

```

After creating the tables, sample data was inserted and the populated relations are shown in the following figures.

The Department Relation

DepartmentID	DepartmentName	College
1	Computer Science	College of Information Technology
2	Information Systems	College of Information Technology
3	Computer Engineering	College of Engineering
4	Electrical Engineering	College of Engineering
5	Mechanical Engineering	College of Engineering
6	Civil Engineering	College of Engineering
7	Mathematics	College of Science
8	Physics	College of Science
9	Chemistry	College of Science
10	Biology	College of Science
11	Business Administration	College of Business Administration
12	Accounting	College of Business Administration
13	Finance	College of Business Administration
14	Economics	College of Business Administration
15	Architecture	College of Engineering
16	Software Engineering	College of Information Technology
17	Cybersecurity	College of Information Technology
18	Data Science	College of Information Technology
19	Renewable Energy En...	College of Engineering
20	Environmental Science	College of Science
*	NULL	NULL

The Researcher Relation

ResearcherID	FullName	Email	Role	DepartmentID
1	Ahmed Ali Hassan	ahmed.hassan@uob.edu.bh	Faculty	1
2	Fatima Salman Mohamed	fatima.salman@uob.edu.bh	Faculty	2
3	Hussain Jassim Alawi	hussain.alawi@uob.edu.bh	Student	1
4	Maryam Abdulrahman Noor	maryam.noor@uob.edu.bh	Student	18
5	Ali Mahdi Abbas	ali.abbas@uob.edu.bh	Faculty	4
6	Zainab Khalil Ahmed	zainab.ahmed@uob.edu.bh	Faculty	17
7	Mohamed Yousif Isa	mohamed.isa@uob.edu.bh	Student	16
8	Noor Hassan Khalifa	noor.khalifa@uob.edu.bh	Faculty	3
9	Sara Faisal Mahmood	sara.mahmood@uob.edu.bh	Student	7
10	Khalid Abdulla Nasser	khalid.nasser@uob.edu.bh	Faculty	11
11	Layla Ebrahim Saleh	layla.saleh@uob.edu.bh	Extern...	20
12	Yusuf Hamad Rashid	yusuf.rashid@uob.edu.bh	Faculty	19
13	Aisha Adel Mansoor	aisha.mansoor@uob.edu.bh	Student	2
14	Sayed Mustafa Almoosawi	mustafa.almoosawi@uob.e...	Faculty	5
15	Mona Sami Darwish	mona.darwish@uob.edu.bh	Faculty	13
16	Abdulla Redha Jawad	abdulla.jawad@uob.edu.bh	Student	8
17	Reem Nabeel Faraj	reem.faraj@uob.edu.bh	Extern...	10
18	Hasan Fadel Amin	hasan.amin@uob.edu.bh	Faculty	6
19	Ruqaya Yaqoob Saeed	ruqaya.saeed@uob.edu.bh	Student	9
20	Omar Adel Kareem	omar.kareem@uob.edu.bh	Faculty	12
*	NULL	NULL	NULL	NULL

The Venue Relation

	VenueID	VenueName	VenueType	Publisher
▶	1	International Journal of Artificial Intelligence Re...	Journal	Springer
	2	GCC Conference on Computing and Technology	Conference	IEEE
	3	Journal of Cybersecurity and Digital Forensics	Journal	Elsevier
	4	Bahrain International Engineering Conference	Conference	University of Bahrain
	5	Renewable Energy Systems Journal	Journal	Taylor and Francis
	6	Middle East Data Science Conference	Conference	ACM
	7	University of Bahrain Research Reports	Report	University of Bahrain
	8	Journal of Information Systems Management	Journal	Emerald
	9	Smart Cities and IoT Symposium	Conference	IEEE
	10	International Journal of Software Engineering	Journal	Springer
	11	Applied Physics Research Journal	Journal	Elsevier
	12	Sustainable Environment Reports	Report	UNESCO
	13	Business Analytics and Finance Conference	Conference	ACM
	14	Journal of Mechanical Design and Manufacturing	Journal	ASME
	15	Civil Infrastructure Research Forum	Conference	ICE Publishing
	16	Book of Advanced Database Systems	Book	Pearson
	17	Journal of Educational Technology	Journal	Taylor and Francis
	18	International Conference on Renewable Energy	Conference	IEEE
	19	Health and Biology Research Journal	Journal	Nature Portfolio
	20	Chemistry and Materials Science Reports	Report	University of Bahrain
*	NULL	NULL	NULL	NULL

The Publication Relation

	PublicationID	Title	PublicationYear	PublicationType	DOI	Abstract	VenueID
▶	1	Artificial Intelligence Applications in Smart Educa...	2024	Journal	10.1000/rpt001	This study explores the use of artificial intelligen...	1
	2	Cybersecurity Awareness Among University Stu...	2023	Conference	10.1000/rpt002	This paper investigates cybersecurity awarenes...	2
	3	Machine Learning for Research Publication Class...	2025	Journal	10.1000/rpt003	This research applies machine learning techniqu...	6
	4	Database Design for Academic Research Tracking	2024	Report	NULL	This report presents a relational database desig...	7
	5	Blockchain-Based Verification of Research Public...	2022	Journal	10.1000/rpt005	This paper discusses blockchain methods for ver...	3
	6	Renewable Energy Adoption in Bahrain Universi...	2023	Conference	10.1000/rpt006	This research studies renewable energy adoptio...	18
	7	Internet of Things Framework for Smart Labora...	2024	Conference	10.1000/rpt007	This paper proposes an IoT framework for moni...	9
	8	Data Mining Techniques for Student Performanc...	2021	Journal	10.1000/rpt008	This study applies data mining methods to analy...	8
	9	Software Engineering Practices in Student Proje...	2025	Journal	10.1000/rpt009	This research evaluates software engineering p...	10
	10	Optimization of Solar Panel Cleaning Systems	2024	Conference	10.1000/rpt010	This paper presents methods to improve solar p...	4
	11	Statistical Modeling in Applied Physics Experiments	2022	Journal	10.1000/rpt011	This study uses statistical modeling to analyze p...	11
	12	Environmental Impact of Plastic Waste Manage...	2023	Report	NULL	This report examines environmental impacts rel...	12
	13	Business Intelligence for Financial Decision Making	2024	Conference	10.1000/rpt013	This paper studies the role of business intelligen...	13
	14	Mechanical Design Improvements for Manufactu...	2021	Journal	10.1000/rpt014	This publication discusses mechanical design me...	14
	15	Civil Infrastructure Risk Assessment Using Data ...	2025	Conference	10.1000/rpt015	This study applies data analytics to civil infrastr...	15
	16	Advanced Relational Database Systems for Res...	2022	Book Chapter	10.1000/rpt016	This chapter explains relational database syste...	16
	17	Digital Learning Platforms and Student Engagem...	2024	Journal	10.1000/rpt017	This research studies the relationship between ...	17
	18	Energy Efficiency in Smart University Buildings	2025	Conference	10.1000/rpt018	This paper examines energy efficiency solutions...	18
	19	Biological Data Analysis Using Machine Learning	2023	Journal	10.1000/rpt019	This study applies machine learning techniques t...	19
	20	Chemical Materials for Sustainable Engineering ...	2022	Report	NULL	This report discusses chemical materials used in ...	20
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL

The Keyword Relation

	KeywordID	KeywordText
▶	12	Applied Physics
	1	Artificial Intelligence
	6	Blockchain
	15	Business Intelligence
	18	Civil Infrastructure
	2	Cybersecurity
	9	Data Mining
	4	Database Systems
	19	Digital Learning
	13	Environmental Sustainability
	16	Financial Analytics
	8	Internet of Things
	3	Machine Learning
	17	Mechanical Design
	14	Plastic Waste
	7	Renewable Energy
	5	Research Tracking
	20	Smart Buildings
	10	Software Engineering
	11	Solar Energy
★	NULL	NULL

The Authorship Relation

	ResearcherID	PublicationID	AuthorOrder
▶	1	1	1
	1	3	2
	1	9	2
	1	16	2
	2	4	1
	2	8	1
	2	16	1
	2	17	2
	3	1	2
	4	3	1
	4	15	2
	5	6	2
	5	14	2
	5	18	2
	6	2	1
	6	5	1
	7	4	2
	7	7	2
	7	9	1
	8	7	1
	8	11	2
	9	8	2
	10	13	1
	11	5	2
	11	12	1

	11	20	2
	12	6	1
	12	10	1
	12	18	1
	13	2	2
	13	17	1
	14	10	2
	14	14	1
	15	13	2
	16	11	1
	17	12	2
	17	19	1
	18	15	1
	19	19	2
	19	20	1
*	NULL	NULL	NULL

The PublicationKeyword Relation

	PublicationID	KeywordID
▶	1	1
	3	1
	17	1
	2	2
	5	2
	1	3
	3	3
	19	3
	4	4
	9	4
	16	4
	3	5
	4	5
	16	5
	5	6
	6	7
	10	7
	18	7
	7	8
	8	9
	11	9
	15	9
	19	9

	9	10
	14	10
	6	11
	10	11
	11	12
	12	13
	20	13
	12	14
	13	15
	13	16
	14	17
	20	17
	15	18
	1	19
	2	19
	8	19
	17	19
	7	20
	18	20
*	NULL	NULL

QUERIES FOR THE SYSTEM:

The following SQL queries were executed on the **Research Publication Tracker** database to retrieve and analyze information from different tables.

Query 1: Publications with Authors, Departments, and Venues

Purpose: This query lists selected publications with their authors, author order, department, and publication venue. It demonstrates a multiple-table join.

```
317 * SELECT
318     p.PublicationID,
319     p.Title,
320     r.FullName AS AuthorName,
321     a.AuthorOrder,
322     d.DepartmentName,
323     v.VenueName
324 FROM Publication p, Authorship a, Researcher r, Department d, Venue v
325 WHERE p.PublicationID = a.PublicationID
326     AND a.ResearcherID = r.ResearcherID
327     AND r.DepartmentID = d.DepartmentID
328     AND p.VenueID = v.VenueID
329     AND p.PublicationID IN (1, 2, 3)
330 ORDER BY p.PublicationID, a.AuthorOrder;
```

PublicationID	Title	AuthorName	AuthorOrder	DepartmentName	VenueName
1	Artificial Intelligence Applications in Smart Educa...	Ahmed Ali Hassan	1	Computer Science	International Journal of Artificial Intelligence Re...
1	Artificial Intelligence Applications in Smart Educa...	Hussain Jassim Alawi	2	Computer Science	International Journal of Artificial Intelligence Re...
2	Cybersecurity Awareness Among University Stu...	Zainab Khali Ahmed	1	Cybersecurity	GCC Conference on Computing and Technology
2	Cybersecurity Awareness Among University Stu...	Aisha Adel Mansoor	2	Information Systems	GCC Conference on Computing and Technology
3	Machine Learning for Research Publication Class...	Maryam Abdulrahman Noor	1	Data Science	Middle East Data Science Conference
3	Machine Learning for Research Publication Class...	Ahmed Ali Hassan	2	Computer Science	Middle East Data Science Conference

Query 2: Publications Newer Than the Average Publication Year

Purpose: This query finds publications that were published after the average publication year in the database. It demonstrates a subquery.

```
333 * SELECT
334     PublicationID,
335     Title,
336     PublicationYear,
337     PublicationType
338 FROM Publication
339 WHERE PublicationYear > (
340     SELECT AVG(PublicationYear)
341     FROM Publication
342 )
343 ORDER BY PublicationYear DESC, PublicationID
344 LIMIT 5;
```

PublicationID	Title	PublicationYear	PublicationType
3	Machine Learning for Research Publication Class...	2025	Journal
9	Software Engineering Practices in Student Proje...	2025	Journal
15	Civil Infrastructure Risk Assessment Using Data ...	2025	Conference
18	Energy Efficiency in Smart University Buildings	2025	Conference
1	Artificial Intelligence Applications in Smart Educa...	2024	Journal

Query 3: Number of Publications per Department

Purpose: This query counts the number of distinct publications connected to each department. It demonstrates GROUP BY.

```
346 * SELECT
347     d.DepartmentName,
348     COUNT(DISTINCT p.PublicationID) AS PublicationCount
349 FROM Department d, Researcher r, Authorship a, Publication p
350 WHERE d.DepartmentID = r.DepartmentID
351     AND r.ResearcherID = a.ResearcherID
352     AND a.PublicationID = p.PublicationID
353 GROUP BY d.DepartmentID, d.DepartmentName
354 ORDER BY PublicationCount DESC, d.DepartmentName
355 LIMIT 5;
```

DepartmentName	PublicationCount
Information Systems	5
Computer Science	4
Electrical Engineering	3
Environmental Science	3
Renewable Energy Engineering	3

Query 4: Latest Publications Ordered by Year and Title

Purpose: This query displays the newest publications first and sorts publications from the same year alphabetically by title. It demonstrates ORDER BY.

```
357 * SELECT
358     PublicationID,
359     Title,
360     PublicationYear,
361     PublicationType
362 FROM Publication
363 ORDER BY PublicationYear DESC, Title ASC
364 LIMIT 5;
```

PublicationID	Title	PublicationYear	PublicationType
15	Civil Infrastructure Risk Assessment Using Data ...	2025	Conference
18	Energy Efficiency in Smart University Buildings	2025	Conference
3	Machine Learning for Research Publication Class...	2025	Journal
9	Software Engineering Practices in Student Proje...	2025	Journal
1	Artificial Intelligence Applications in Smart Educa...	2024	Journal

Query 5: Researchers Who Have Not Authored Any Publication

Purpose: This query finds researchers who do not appear in the Authorship table. It demonstrates NOT IN.

```
366 * SELECT
367     r.ResearcherID,
368     r.FullName,
369     d.DepartmentName
370 FROM Researcher r, Department d
371 WHERE r.DepartmentID = d.DepartmentID
372 AND r.ResearcherID NOT IN (
373     SELECT ResearcherID
374     FROM Authorship
375 )
376 ORDER BY r.FullName;
```

ResearcherID	FullName	DepartmentName
20	Omar Adel Kareem	Accounting

Query 6: Publications Related to Energy

Purpose: This query searches for publications that contain the word "Energy" in the title or abstract. It demonstrates LIKE.

```
378 * SELECT
379     PublicationID,
380     Title,
381     PublicationYear,
382     PublicationType
383 FROM Publication
384 WHERE Title LIKE 'Energy%'
385 OR Abstract LIKE 'Energy%'
386 ORDER BY PublicationYear DESC, Title;
```

PublicationID	Title	PublicationYear	PublicationType
18	Energy Efficiency in Smart University Buildings	2025	Conference
6	Renewable Energy Adoption in Bahrain Univers...	2023	Conference

Query 7: Overall Publication Year Statistics

Purpose: This query summarizes the publication years in the database. It demonstrates aggregate functions such as COUNT, MIN, MAX, and AVG.

```
388 * SELECT
389     COUNT(*) AS TotalPublications,
390     MIN(PublicationYear) AS EarliestYear,
391     MAX(PublicationYear) AS LatestYear,
392     ROUND(AVG(PublicationYear), 1) AS AverageYear
393 FROM Publication;
```

TotalPublications	EarliestYear	LatestYear	AverageYear
20	2021	2025	2023.3

Query 8: Researchers with More Than Two Publications

Purpose: This query lists researchers who authored more than two publications. It demonstrates HAVING.

```
395 * SELECT
396     r.ResearcherID,
397     r.FullName,
398     COUNT(a.PublicationID) AS PublicationCount
399 FROM Researcher r, Authorship a
400 WHERE r.ResearcherID = a.ResearcherID
401 GROUP BY r.ResearcherID, r.FullName
402 HAVING COUNT(a.PublicationID) > 2
403 ORDER BY PublicationCount DESC, r.FullName;
```

ResearcherID	FullName	PublicationCount
1	Ahmed Ali Hassan	4
2	Fatima Salman Mohamed	4
5	Ali Mahdi Abbas	3
11	Layla Ebrahim Saleh	3
7	Mohamed Yousif Isa	3
12	Yusuf Hamad Rashid	3

Query 9: Publication Age in Years

Purpose: This query calculates the age of each publication using the project year 2026. It demonstrates a calculated field.

```
405 * SELECT
406     PublicationID,
407     Title,
408     PublicationYear,
409     (2026 - PublicationYear) AS PublicationAgeYears
410 FROM Publication
411 ORDER BY PublicationAgeYears DESC, PublicationID
412 LIMIT 5;
```

PublicationID	Title	PublicationYear	PublicationAgeYears
8	Data Mining Techniques for Student Performanc...	2021	5
14	Mechanical Design Improvements for Manufactu...	2021	5
5	Blockchain-Based Verification of Research Public...	2022	4
11	Statistical Modeling in Applied Physics Experiments	2022	4
16	Advanced Relational Database Systems for Res...	2022	4

Query 10: Complete Publication Report

Purpose: This query produces a detailed publication report showing publication, venue, authors, departments, and keywords. It involves most of the database relations.

```
414 * SELECT
415     p.PublicationID,
416     p.Title,
417     p.PublicationYear,
418     v.VenueName,
419     GROUP_CONCAT(DISTINCT r.FullName ORDER BY a.AuthorOrder SEPARATOR ', ') AS Authors,
420     GROUP_CONCAT(DISTINCT d.DepartmentName ORDER BY d.DepartmentName SEPARATOR ', ') AS Departments,
421     GROUP_CONCAT(DISTINCT k.KeywordText ORDER BY k.KeywordText SEPARATOR ', ') AS Keywords
422 FROM Department d, Researcher r, Authorship a, Publication p, Venue v, PublicationKeyword pk, Keyword k
423 WHERE d.DepartmentID = r.DepartmentID
424     AND r.ResearcherID = a.ResearcherID
425     AND a.PublicationID = p.PublicationID
426     AND p.VenueID = v.VenueID
427     AND p.PublicationID = pk.PublicationID
428     AND pk.KeywordID = k.KeywordID
429 GROUP BY p.PublicationID, p.Title, p.PublicationYear, v.VenueName
430 ORDER BY p.PublicationID
431 LIMIT 5;
```

PublicationID	Title	PublicationYear	VenueName	Authors	Departments
1	Artificial Intelligence Applications in Smart Educa...	2024	International Journal of Artificial Intelligence Re...	Ahmed Ali Hassan, Hussain Jassim Alawi	Computer Science
2	Cybersecurity Awareness Among University Stu...	2023	GCC Conference on Computing and Technology	Zainab Khalil Ahmed, Aisha Adel Mansoor	Cybersecurity, Inf
3	Machine Learning for Research Publication Class...	2025	Middle East Data Science Conference	Maryam Abdulrahman Noor, Ahmed Ali Hassan	Computer Science,
4	Database Design for Academic Research Trading	2024	University of Bahrain Research Reports	Fatima Salman Mohamed, Mohamed Yousif Isa	Information System
5	Blockchain-Based Verification of Research Public...	2022	Journal of Cybersecurity and Digital Forensics	Zainab Khalil Ahmed, Layla Ebrahim Saleh	Cybersecurity, Em

CONTRIBUTIONS:

MEETING NO.1	8 TH MAY 2026	Discussed the project idea, identified the main system requirements, and planned the database entities and relationships.
MEETING NO.2	10 TH MAY 2026	Designed the ER diagram and converted it into the relational schema with primary keys and foreign keys.
MEETING NO.3	14 TH MAY 2026	Implemented the database in MySQL, created the tables using SQL, inserted sample data, and tested SQL queries.
MEETING NO.4	15 TH MAY 2026	Reviewed the project report, organized screenshots and query results, checked formatting, and prepared the final submission.

Member Name	Contribution
Hussain Ali H. Ali	Database implementation, SQL queries, sample data insertion
Dania Ahmed	Report writing, formatting, ER diagram organization, testing queries

CONCLUSION:

The **Research Publication Tracker database system** was successfully designed and implemented using **MySQL**. The project applied important database concepts such as ER modeling, relational schema design, constraints, normalization, and SQL query processing. The system allows efficient storage and retrieval of publication-related information and demonstrates the practical use of database management systems in academic environments.

END OF REPORT
